Software Requirements Specification — Online Examination Portal

Course: UE24CS341A — Software Engineering, PES University, Dept. of CSE
Project: Online Examination System with Multiple-Choice Questions and Evaluation
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Version: 4.2 · **Date:** 2026-11-12 · **Status:** Baselined; all requirements implemented on `main`

Revision history

Version	Date	Author	Change summary	Approval
M1	2026-08-19	Team	First requirements baseline: 83 functional and 27 non-functional requirements	—
M1 rev.	2026-09-06	Kshitij	Scope cut by CR-01 (decided 2 Sep) to 30 FRs and 10 NFRs; IDs kept, dropped IDs retired	—
3.0	2026-10-15	Kshitij	Rebuilt to the Part-2 deliverables; security requirements NFR-011 … NFR-014 added from the security analysis	—
3.1	2026-11-03	Kshitij	Arranged to IEEE 830: Overall Description, use case diagram, security objectives SO-1 … SO-3. No requirement changed	PR #111
4.0	2026-11-11	Kshitij	Arranged to the course SRS template: External Interfaces (§3), test case references on every requirement (§4, §5), Quality Attributes (§6), two use case diagrams (§7) and the RTM (§8). No requirement added, removed or changed	Pending review
4.1	2026-11-12	Kshitij	NFR-001 and NFR-003 verified by the 200-user load test (Deviations D-18): RTM status N → P. No requirement changed	PR #114
4.2	2026-11-12	Kshitij	CR-02, the instructor's rule of 30 Oct: Faculty send a finished paper to the Administrator, who schedules and so publishes it, or returns it with a note. FR-012 and FR-013 reworded; definitions, roles and use case diagram 2 updated (Deviations D-20)	Pending review

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			
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1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements of the Online Examination Portal, with acceptance criteria and the test cases that verify each one.

It is the single source for the GitHub backlog (one item per requirement), the Use Case Diagram and Flow, the Software Architecture Document (`docs/SAD.pdf`) and the Test Plan (`docs/Test_Plan.pdf`).

Related document	Where
IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications	—
Use Case Diagram and Use Case Flow (UC-01 … UC-13)	`docs/Use_Case_Diagram.pdf`, `docs/Use_Case_Flow.pdf`
Software Architecture and Design Specification	`docs/SAD.pdf`
Software Test Plan (119 cases)	`docs/Test_Plan.pdf`
Sprint 1 and Sprint 2 plan	`docs/Sprint_Plan.pdf`
Deviations register	`docs/Deviations.pdf`
Backlog — one issue per FR / NFR, with story points, assignee and sprint	GitHub Project *Online Exam Portal - Backlog*

1.2 Scope

Faculty author multiple-choice questions into a tagged question bank and assemble them into an examination, which the Administrator schedules for a fixed window.

Enrolled Students attempt the examination under a server-enforced timer with autosave and resume. Every submission is evaluated automatically against the stored answer key. Results are released when Faculty publishes them.

Out of scope: descriptive answers and manual grading · webcam or AI proctoring · biometric identity verification · payments · adaptive testing · native mobile applications · offline mode · plagiarism detection.

Why 30 functional requirements. The M1 baseline (19 Aug) held 83 functional and 27 non-functional requirements. That is more than a team of three can build, test at 8–10 cases per use case, and demonstrate in two one-week sprints.

On 2 Sep it was cut to the demonstration-critical core. Every requirement needed to take one examination through its whole lifecycle (set up → attempt → evaluate → publish) was kept.

Conveniences such as automatic paper generation, CSV import and export, multiple-correct questions and per-student shuffling were dropped. The four security requirements NFR-011 … NFR-014 were added on 15 Oct from the project's security analysis.

1.3 Audience

Reader	Uses this document to
Developers (the team)	Build each requirement to its acceptance criteria
QA (the team)	Derive and run the test cases referenced against each requirement
Course instructor and evaluators	Assess the requirements, their verification and their traceability
Maintainers	Find which component realises a requirement (§8) and why it exists

1.4 Definitions

Term	Meaning
Examination	A paper of single-correct MCQs belonging to one Course, with its own duration, marks and window
Window	The start and end date-time between which an Examination can be started
Attempt	One Student's sitting of one Examination, stamped with the server start time
Send to Admin	Faculty handing a finished Draft to the Administrator for scheduling (Draft → Sent to Admin); the paper is then locked (CR-02)
Publish	The Administrator releasing an Examination to Students by scheduling it (Sent to Admin → Published), or Faculty releasing its results
Focus loss	The examination tab or window losing focus during an active attempt
FR / NFR / UC	Functional requirement / non-functional requirement / use case
SO	Security objective (§5.1.1)
AC	Acceptance criterion; `AC-FR-001` is the criterion of FR-001
UT / IT / ST	Unit / integration / system test case of the Test Plan, e.g. `UT_001`
RTM	Requirements Traceability Matrix (§8)
DPDP Act	India's Digital Personal Data Protection Act, 2023
WCAG	Web Content Accessibility Guidelines 2.1
CSRF	Cross-site request forgery
ORM	Object-relational mapper — Django's database layer
HTTPS / TLS	HTTP over Transport Layer Security

2. Overall description

2.1 Product perspective

The portal is a new, self-contained web application: one Django server that browsers reach over HTTPS, with its own database.

It replaces paper MCQ tests and manual marking for a department and has no interface to other university systems. `docs/SAD.pdf` describes its architecture.

2.2 Major product functions

An examination goes through one lifecycle, and every functional requirement serves a step of it:

9. **Set up** — Administrators create Courses and enroll Students; Faculty write questions into a tagged bank and assemble them into an Examination, which the Administrator schedules (FR-004 … FR-013).
10. **Attempt** — an enrolled Student starts the Examination inside its window and answers one question at a time under a server timer, with every answer saved as it is chosen (FR-014 … FR-021).
11. **Evaluate** — every submission, manual or on time-out, is scored at once against the answer key (FR-022 … FR-024).
12. **Publish** — Faculty release the results, and each Student sees a scorecard (FR-025 … FR-027).

Across the lifecycle, accounts and roles (FR-001 … FR-003) control who can do what. The integrity requirements (FR-028 … FR-030) record focus loss and every change.

2.3 User roles and characteristics

Actor	Description
Student	An enrolled candidate who sits examinations and views published results. Uses the portal a few times a term, on a personal device
Faculty	Authors questions and examinations, sends them to the Administrator for scheduling, publishes results. Uses the portal throughout the term
Administrator	Manages courses, enrolls students, schedules (and so publishes) the examinations Faculty send, and reviews the audit trail
Clock	The server clock — triggers automatic submission when an attempt's time runs out

2.4 Operating environment

The server runs on one Linux virtual machine (Django 5.2, Python 3.12, SQLite). Clients use the two latest versions of Chrome, Edge or Firefox at a viewport of 360 px or wider (NFR-005).

2.5 Constraints

- One Git repository, feature branches and reviewed pull requests (NFR-006).
- Secrets only from environment variables, because the repository is public (NFR-014).
- Every timing rule reads the server clock, never the client's (FR-017).

- Personal data handled under the DPDP Act (NFR-008); the attempt flow meets WCAG 2.1 AA (NFR-009).
- Two one-week sprints and a team of three, which bounded the requirement set (§1.2).

2.6 Assumptions, dependencies and open items

Assumptions and dependencies.

- Students have a personal device with a supported browser and an internet connection.
- Examinations are attempted remotely.
- The server clock is synchronised by NTP, since every timing rule depends on it.
- The system is one Django application with an SQLite database on one Linux virtual machine.

Open items awaiting the course instructor. The assumption taken is used throughout this SRS and the Test Plan:

Open question	Affects	Assumption taken
Negative marking — a flat deduction per wrong answer, or a fraction of the question's marks	FR-023, UC-11	**Flat** deduction of the question's configured negative marks
Is a score exactly at the pass mark a Pass?	FR-024, UC-11	**Inclusive** — 40.0% against a 40% pass mark is a Pass

3. External interface requirements

3.1 User interfaces

- **Server-rendered web pages** in the browser. Each role lands on its own dashboard after sign-in (FR-002): Student, Faculty or Administrator.
- **Examination screen:** one question at a time, a navigation palette showing each question's state (FR-018), and a countdown computed by the server (FR-017).
- **Forms** report invalid input next to the field that caused it. Refusals name the reason in the user's terms ("This examination has not opened yet").
- **Errors** show one generic page with no stack trace or internal detail (NFR-013).
- **Accessibility:** the attempt flow is keyboard-operable, has a skip link to the main content, labels every option for screen readers and keeps text contrast at 4.5:1 or more (NFR-009).
- **Small screens:** every page is usable at 360 px wide without horizontal scrolling (NFR-005).

3.2 Hardware interfaces

The system needs no special hardware. A client is any device that runs a supported browser, used with a keyboard, mouse or touch screen. The server is one Linux virtual machine.

No camera, microphone or biometric device is used: proctoring and biometric checks are out of scope (§1.2).

3.3 Software interfaces

Interface	With	Purpose
HTML pages and form posts	The browser	Every screen; forms use Post/Redirect/Get
JSON endpoints under `/api/`	The page scripts in the browser	Autosave, remaining time and focus-loss reports (`docs/SAD.pdf` §4.3)
Django ORM	SQLite database	All data access; no SQL is built by hand (NFR-013)
Environment variables	Host operating system	`DJANGO_SECRET_KEY` and other secrets (NFR-014)
NTP	Host clock	Keeps the server clock right, on which every timing rule depends
GitHub Actions	Repository	Runs the full test suite on every push and pull request

There is no interface to any other university system (§2.1).

3.4 Communications

- **HTTPS** between browser and server, terminated by Nginx in the deployment of `docs/SAD.pdf` §3.6. The validation runs used the development server on `localhost`.
- Every state-changing request carries an **anti-CSRF token** (NFR-013). The session identifier is regenerated at sign-in (IT_003).
- **Autosave** sends each response as soon as it is chosen and must reach the server within 5 s (FR-019). A save is shown as Answered only after the server confirms it.
- An unauthenticated call to `/api/` gets `401` JSON; an unauthenticated page request is redirected to sign-in (`302`).

4. System features (detailed)

Each requirement below carries an acceptance criterion and the Test Plan cases that verify it. Functional IDs follow `FR-###`; "Tests" lists the cases of `docs/Test_Plan.pdf`.

Against the course minimum: 30 functional requirements (at least 15), 14 non-functional (at least 5), 3 security objectives (at least 2) and 10 security requirements (at least 5).

Comments name the use case, the sprint that built the requirement, how it is verified, why it exists, and the requirements it depends on.

4.1 User Management & Authentication

Description: let Students register, let everyone sign in, and confine each account to one role.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
FR-001	The system shall allow a visitor to register a Student account by submitting full name, institutional email, roll number and password.	Functional	High	Student	AC-FR-001: a submission with all four valid fields creates a Student account; a missing field, a non-institutional email, or a duplicate email or roll number is rejected with an inline error and no account is created. Tests: UT_001–UT_006, IT_001, IT_002, ST_001	UC-01 · Sprint 1 · Functional test. Why: Students must be able to join without waiting for an administrator

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
FR-002	The system shall authenticate a user against email and password and establish a session on success.	Functional	High	All users	AC-FR-002: correct credentials open the dashboard for the user's role; a wrong password shows "Invalid email or password" and no session is created. Tests: UT_007–UT_010, IT_003, ST_002	UC-02 · Sprint 1 · Functional test. Why: every other function depends on a known, signed-in user
FR-003	The system shall assign exactly one role — Student, Faculty or Administrator — to every account, and shall restrict every screen and endpoint to the roles authorised for it.	Functional	High	Security	AC-FR-003: a Student requesting a Faculty URL directly receives HTTP 403; an unauthenticated request to any protected URL is redirected to sign-in (HTTP 302) or refused (HTTP 401). Tests: UT_011, IT_004–IT_006, ST_002	UC-02 · Sprint 1 · Security test. Why: hiding a button is not access control; the server must refuse. Depends on FR-002

4.2 Course & Enrollment Management

Description: Courses group everything else; enrollment decides who may sit which paper.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
FR-004	The system shall allow an Administrator to create, edit and archive a Course identified by course code, title and semester, and to assign one or more Faculty accounts to it.	Functional	High	Administrator	AC-FR-004: a Course created with code, title and semester appears in the course list with its assigned Faculty; a duplicate course code is rejected; an archived Course accepts no new examinations. Tests: UT_012–UT_016, IT_007, IT_008, ST_003	UC-03 · Sprint 1 · Functional test. Why: examinations, questions and enrollments all hang off a Course
FR-005	The system shall allow Faculty or an Administrator to enroll Student accounts into a Course individually.	Functional	High	Faculty, Administrator	AC-FR-005: after enrolling a Student, that Student appears on the Course roster; enrolling the same Student twice leaves exactly one enrollment. Tests: UT_017–UT_020, ST_004	UC-04 · Sprint 1 · Functional test. Why: enrollment decides who may sit which examination. Depends on FR-004
FR-006	The system shall permit a Student to view and attempt only those examinations that belong to Courses in which the Student is enrolled.	Functional	High	Security, Faculty	AC-FR-006: a Student not enrolled in the Course does not see its examination and receives HTTP 403 on a direct URL to it. Tests: UT_021, IT_009–IT_011, ST_004	UC-04 · Sprint 1 · Integration test. Why: stops students sitting papers of courses they are not registered for. Depends on FR-005

4.3 Question Bank Management

Description: Faculty write single-correct MCQs into a tagged bank that keeps past papers intact.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
FR-007	The system shall allow Faculty to create a single-correct multiple-choice question consisting of a stem, between 2 and 6	Functional	High	Faculty	AC-FR-007: a question with 4 options and one correct option is saved; 1 option, 7 options, zero correct or two correct options	UC-05 · Sprint 1 · Functional test. Why: MCQ authoring is the core content of the system

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
	options, and exactly one option marked correct.				are each rejected with an error. Tests: UT_022–UT_026, ST_005	
FR-008	The system shall require every question to carry a Course, a topic label, a difficulty of Easy, Medium or Hard, positive marks, and negative marks.	Functional	High	Faculty	AC-FR-008: saving a question with any of the five fields missing, a difficulty outside the three values, or negative positive-marks is rejected. Tests: UT_027, IT_012, ST_005	UC-05 · Sprint 1 · Data validation test. Why: tags make the bank searchable; marks drive scoring. Depends on FR-007
FR-009	The system shall prevent modification or deletion of a question that belongs to a published or attempted examination, and shall instead create a new version of that question.	Functional	Medium	Faculty, Integrity	AC-FR-009: editing a question used in a published examination creates version 2; the examination still shows version 1 unchanged; deleting it is refused. Tests: IT_013, IT_014, ST_005	UC-05 · Sprint 2 · Database integrity test. Why: changing a question under a live paper would make past scores wrong. Depends on FR-007, FR-012

4.4 Examination Authoring

Description: Faculty build a paper from the bank, set its rules and publish it when ready.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
FR-010	The system shall allow Faculty to create an Examination with a title, Course, duration in minutes, declared total marks, instructions, and the policy values: negative marking (on/off and value), question and option shuffling, and pass mark.	Functional	High	Faculty	AC-FR-010: an Examination saved with all fields is listed in Draft state with the entered values; a duration of 0 or a pass mark above 100% is rejected. Tests: UT_028–UT_031, ST_006	UC-06 · Sprint 1 · Functional test. Why: each paper needs its own rules. Depends on FR-004
FR-011	The system shall allow Faculty to add questions to an Examination by selecting them from the question bank.	Functional	High	Faculty	AC-FR-011: questions selected from the bank appear in the Examination in the selected order; a question from another Course cannot be added. Tests: IT_015, ST_006	UC-06 · Sprint 1 · Functional test. Why: papers are built by reuse, not retyping. Depends on FR-007, FR-010
FR-012	The system shall hold an Examination in Draft state until Faculty sends it to the Administrator, shall lock it while it waits to be scheduled, shall let the Administrator return it to Draft with a note, and shall make only published Examinations visible to Students.	Functional	High	Faculty, Administrator (CR-02)	AC-FR-012: a Draft or Sent-to-Admin examination is not visible on any Student dashboard and a sent one cannot be edited; once the Administrator schedules it, it is visible to enrolled Students; a returned paper is a Draft again and shows the note. Tests: UT_032, UT_033, IT_016, ST_006	UC-06 · Sprint 1; Faculty → Administrator hand-off by CR-02 (Deviations D-20) · State transition test. Why: Faculty must be able to prepare a paper privately. Depends on FR-010

4.5 Scheduling & Access Control

Description: the Administrator gives a sent paper its window, which publishes it; it then appears on the Student's dashboard and can be started only inside that window, once.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
FR-013	The system shall allow the Administrator to schedule an Examination that Faculty has sent by setting a window start date-time and end date-time; scheduling it publishes it.	Functional	High	Administrator (CR-02)	AC-FR-013: the window is saved and shown on the Examination; an end time earlier than the start time is rejected. Tests: UT_034, UT_035, IT_017, ST_007	UC-07 · Sprint 1 · Functional test. Why: the window decides when the paper can be taken. Depends on FR-012
FR-014	The system shall present to each Student a dashboard listing the Upcoming, Live and Completed Examinations of the Courses in which the Student is enrolled.	Functional	High	Student	AC-FR-014: with server time before, inside and after the window, the examination appears under Upcoming, Live and Completed respectively. Tests: UT_036–UT_038, IT_018, ST_007	UC-07 · Sprint 2 · Functional test. Why: Students need one place to find their papers. Depends on FR-006, FR-013
FR-015	The system shall permit a Student to begin an attempt only when the current server time lies within the window, the Student is enrolled, and the Student has no previously submitted attempt for that Examination.	Functional	High	Faculty (fairness)	AC-FR-015: Start is refused with a message before the window opens, after it closes, for a non-enrolled Student and for a second attempt; it succeeds otherwise. Tests: UT_039–UT_041, ST_008	UC-08 · Sprint 1 · System test. Why: same window, one attempt each. Depends on FR-005, FR-013

4.6 Examination Delivery (Attempt)

Description: deliver the paper under a server-held timer, save every answer at once, and survive interruptions without adding time.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
FR-016	The system shall create an attempt record stamped with the server start time when a Student begins an Examination, and shall deliver the paper only after that record exists.	Functional	High	Integrity	AC-FR-016: the attempt row with a server start time exists before the first question is returned; no question is returned if the row cannot be created. Tests: IT_019, IT_020, ST_008	UC-08 · Sprint 1 · Sequence test. Why: no one can see the paper without a timed attempt on record. Depends on FR-015
FR-017	The system shall display a countdown timer whose remaining time is derived from the server clock and the attempt start time, and shall not rely on the client clock.	Functional	High	Security	AC-FR-017: moving the client PC clock forward by 30 minutes does not change the remaining time shown after a page reload. Tests: UT_042, ST_008	UC-08 · Sprint 1 · Security / unit test. Why: a client clock can be tampered with. Depends on FR-016
FR-018	The system shall display one question at a time with a navigation palette showing each question as Not Visited, Not Answered, Answered or Marked for Review.	Functional	High	Student	AC-FR-018: answering, skipping and marking questions changes each palette cell to the matching state, and the states survive a page reload. Tests: UT_043, UT_044, IT_021, ST_008	UC-08 · Sprint 2 · UI / state test. Why: lets students pace and review a long paper. Depends on FR-016
FR-019	The system shall persist every response to the server within 5 seconds of the Student selecting it, without any explicit save action.	Functional	High	Student	AC-FR-019: 5 seconds after selecting an option, the response is present in the database; killing the browser then loses no response older than 5 s. Tests: UT_045–UT_047, IT_022, IT_023, ST_009	UC-09 · Sprint 2 · Resilience / API test. Why: a dropped connection must not cost a student their answers. Depends on FR-016

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
FR-020	The system shall automatically submit an attempt, retaining all responses persisted up to that instant, when the remaining time reaches zero.	Functional	High	Faculty (fairness)	AC-FR-020: an attempt left open past its duration is in Submitted state with all saved responses, and no response saved after expiry is accepted. Tests: UT_049–UT_052, IT_026–IT_028, ST_010	UC-10 · Sprint 1 · Automation test. Why: time limits must be enforced the same for every student. Depends on FR-017, FR-019
FR-021	The system shall allow a Student whose session was interrupted to resume the same attempt, with remaining time computed from the original server start time and all saved responses restored.	Functional	High	Student	AC-FR-021: closing the browser 10 minutes into a 60-minute attempt and reopening it 5 minutes later shows about 45 minutes left and every earlier answer. Tests: UT_048, IT_024, IT_025, ST_009	UC-09 · Sprint 2 · Fault-injection test. Why: interruptions must neither erase work nor add time. Depends on FR-016, FR-019

4.7 Automatic Evaluation & Scoring

Description: score every submission at once, exactly, with no human step.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
FR-022	The system shall evaluate every submitted attempt automatically, without any human intervention.	Functional	High	Faculty	AC-FR-022: every attempt in Submitted state, whether submitted manually or automatically, has a stored score without any Faculty action. Tests: IT_028–IT_030, ST_011	UC-11 · Sprint 1 · Unit test. Why: removes manual grading entirely. Depends on FR-020
FR-023	The system shall award the question's full positive marks for a correct response, deduct the configured negative marks for an incorrect response, and award zero with no deduction for an unanswered question.	Functional	High	Faculty, Department	AC-FR-023: with +2 / −0.5 marking, 10 correct, 4 wrong and 6 unanswered scores exactly 18.0. Tests: UT_053–UT_056, IT_031, ST_011	UC-11 · Sprint 1 · Calculation test. Why: the scoring rule must be exact and predictable. Depends on FR-008, FR-010
FR-024	The system shall compute, for every evaluated attempt, the total score, the percentage and a Pass or Fail outcome against the Examination's pass mark.	Functional	High	Student, Faculty	AC-FR-024: 20 out of 50 with a 40% pass mark is stored as 40.0% and Pass; 19.5 out of 50 is 39.0% and Fail. Tests: UT_057, UT_058, IT_031, ST_011	UC-11 · Sprint 2 · Calculation test. Why: Students and Faculty need the outcome, not only the raw score. Depends on FR-023

4.8 Results

Description: results stay hidden until Faculty release them; the scorecard shows how each mark was earned.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
FR-025	The system shall withhold every evaluated result from the Student until Faculty publishes the results of that Examination.	Functional	High	Faculty	AC-FR-025: before publication the Student sees only "Submitted" with no score, and the results URL returns HTTP 403. Tests: UT_059, IT_032, IT_033, ST_012	UC-12 · Sprint 2 · Access control test. Why: Faculty decide when results are released. Depends on FR-022

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
FR-026	The system shall allow Faculty to publish and to unpublish the results of an Examination.	Functional	High	Faculty	AC-FR-026: after Publish the scorecard is visible to the Student; after Unpublish it is hidden again. Tests: UT_060, UT_061, IT_034, ST_012	UC-12 · Sprint 2 · Functional test. Why: Faculty need to correct a premature release. Depends on FR-025
FR-027	The system shall display to a Student, for a published result, the total score, the percentage, the Pass or Fail outcome and the marks obtained on each question.	Functional	High	Student	AC-FR-027: every figure on the scorecard equals the stored evaluation exactly, including per-question marks. Tests: UT_062, UT_063, IT_034, ST_012	UC-12 · Sprint 2 · Functional test. Why: Students should see how the score was reached. Depends on FR-024, FR-026

4.9 Integrity & Audit

Description: give Faculty evidence of focus loss, keep the answer key off the client, and record every change.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
FR-028	The system shall record, with a server-side timestamp, every occurrence of the examination window losing focus during an active attempt.	Functional	Medium	Faculty	AC-FR-028: switching tabs three times during an attempt creates three focus-loss records with server timestamps, visible to Faculty. Tests: UT_064–UT_067, ST_013	UC-13 · Sprint 2 · Event capture test. Why: gives Faculty evidence; a human decides. Depends on FR-016
FR-029	The system shall exclude all correct-answer data from every response delivered to a Student during an active attempt.	Functional	High	Security	AC-FR-029: the page source and every network response of an attempt contain no correct-option field or value. Tests: IT_035, ST_013	UC-13 · Sprint 1 · Payload inspection. Why: leaking the key in the page would void the exam. Depends on FR-016
FR-030	The system shall write an append-only audit entry recording the actor, action, affected entity, timestamp and changed values for every creation, modification and deletion of a user, question, examination, answer key or result.	Functional	Medium	Administrator	AC-FR-030: editing a question creates one audit row with old and new values; no screen or endpoint can edit or delete an audit row. Tests: UT_068, IT_036, IT_037, ST_013	UC-13 · Sprint 2 · Security audit. Why: disputes need an untampered record of who changed what

Functional requirement count: 30 — Sprint 1: 19 · Sprint 2: 11.

5. Non-functional requirements (detailed)

NFRs are measurable and tied to the Test Plan. IDs follow NFR-###. Each is classed as **Product**, **Organisational** or **External**, with its quality category in brackets.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement	Comments
NFR-001	The system shall complete 95% of page loads and requests within 2 seconds while 200 attempts are	Product (Performance)	High	A Locust load test with 200 simulated students shows a 95th-percentile response time ≤ 2 s. Test:	Sprint 2. Why: a slow paper costs students time

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement	Comments
	active concurrently.			`loadtest/locustfile.py`	
NFR-002	The system shall evaluate a submitted 100-question attempt within 5 seconds.	Product (Performance)	High	A timed test of evaluating a 100-question attempt completes in ≤ 5 s. Test: IT_030 (`evaluation/test_performance.py`)	Sprint 2. Why: results must be ready right after the exam
NFR-003	The system shall sustain at least 200 concurrent attempts and a question bank of at least 50,000 questions without breaching NFR-001.	Product (Capacity)	Medium	NFR-001's load test passes with the bank seeded to 50,000 questions. Test: `loadtest/locustfile.py`	Sprint 2. Why: a department-wide exam must not degrade the system
NFR-004	After an unplanned server restart, the system shall restore service within 60 seconds, with every in-progress attempt resumable with its saved responses and correct remaining time.	Product (Reliability)	High	Killing and restarting the server during an attempt restores service in ≤ 60 s and the attempt resumes intact. Test: `portal/test_restart.py`	Sprint 2. Why: a crash must not void an examination
NFR-005	The system shall work on the two latest versions of Chrome, Edge and Firefox, and remain usable at viewport widths of 360 px and above.	Product (Portability)	Medium	The attempt flow completes on each browser and at 360 px width without horizontal scrolling. Test: `portal/test_responsive.py`; browser check in the System Validation Report	Sprint 2. Why: Students use their own devices
NFR-006	All code and documents shall be held in one Git repository; work shall proceed on feature branches; every merge to `main` shall be through a pull request raised by the assigned member and reviewed by another member.	Organisational	High	The GitHub history shows no direct commits to `main` after 19 Oct, and every merged PR has a reviewer other than its author. Test: review of the GitHub history	Sprint 1. Why: required by the course; keeps `main` working
NFR-007	Every functional requirement shall be linked to at least one test case in the Test Plan.	Organisational	Medium	The coverage table of `docs/Test_Plan.pdf` lists all 30 FRs. Test: `portal/test_traceability.py`	Sprint 2. Why: untested requirements are unverified claims
NFR-008	Personal data shall be handled per the Digital Personal Data Protection Act, 2023: consent at registration, collection only for examination use, and erasure on request within 30 days.	External (Legal)	Low	Registration cannot complete without ticking consent; an Administrator can erase a Student's personal data. Test: `identity/test_dpdp.py`	Sprint 2. Why: Indian law applies to student data
NFR-009	The attempt flow shall meet WCAG 2.1 Level AA: fully keyboard-operable, text contrast at least 4.5:1, and every option control labelled for screen readers.	External (Accessibility)	Medium	An axe/Lighthouse audit reports no AA violations, and the whole attempt can be completed with the keyboard alone. Test: `portal/test_accessibility.py`; axe audit in the System Validation Report	Sprint 2. Why: every student must be able to sit the exam
NFR-010	Examination policy values — window, pass mark and negative marking — shall be configurable per examination and shall not be hard-coded.	External (Maintainability)	Medium	Code review finds no hard-coded pass mark or negative-marking value; changing them on an examination changes its scoring. Test: `evaluation/test_policy.py`	Sprint 2. Why: departments use different rules
NFR-011	Passwords shall be stored only as salted hashes using an adaptive algorithm, and shall never be stored or logged in plain text.	Product (Security)	High	The password column holds only `pbkdf2_sha256$…` hashes, and the logs contain no password after a registration and a sign-in. Test: IT_001 (`identity/test_auth.py`)	Sprint 1. Why: a database leak must not leak passwords
NFR-012	The system shall lock an account for 15 minutes after 5 failed sign-in attempts within 15 minutes.	Product (Security)	Medium	The 6th sign-in attempt after 5 failures is refused even with the right password, and succeeds after 15 minutes. Test: IT_038	Sprint 2. Why: blocks password guessing

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement	Comments
				(`identity/test_lockout.py`)	
NFR-013	Every state-changing request shall carry an anti-CSRF token, all database access shall go through the ORM (no string-built SQL), and error pages shall show no stack trace or internal detail.	Product (Security)	High	A POST without a CSRF token is refused with HTTP 403; a code search finds no raw SQL; a forced error shows a generic page with DEBUG=`False`. Test: `portal/test_security.py`	Sprint 1. Why: covers the most common web attacks
NFR-014	Secrets such as the Django SECRET_KEY and database credentials shall not be committed to the repository; they shall be read from environment variables.	Product (Security)	High	A secret scan of the public repository finds no secret, and the app refuses to start without the SECRET_KEY variable. Test: secret scan and start-up check in the System Validation Report	Sprint 1. Why: the repository is public

Non-functional requirement count: 14 — Sprint 1: 4 · Sprint 2: 10.

5.1 Security

5.1.1 Security objectives

The examination is only worth something if its results can be trusted. The security work serves three objectives:

ID	Objective	Threat it answers
SO-1	**Confidentiality of the examination** — no Student sees the answer key, another Student's attempt, or a result before it is published	A key in the page source; guessing another Student's URL
SO-2	**Integrity of attempts and scores** — time limits, scores and the record of changes cannot be altered from the browser	Changing the PC clock; forged requests; quietly editing a question after the exam
SO-3	**Protection of accounts and secrets** — credentials and keys stay safe even if the database or the repository is exposed	A leaked database; password guessing; a key committed to the public repository

5.1.2 Security requirements

The ten security requirements keep their FR / NFR IDs, so the SRS, Test Plan, RTM and test code share one ID space. Each is specified in full in §4 or §5.

How each is validated is in §5.1 of `docs/Test_Plan.pdf`. How each is built is in §3.9 of `docs/SAD.pdf`.

Req ID	Requirement (shall...)	Type	Priority	Objective	Acceptance criteria / Test case ref
FR-003	The system shall restrict every screen and endpoint to the roles authorised for it	Security (access control)	High	SO-1, SO-2	Another role's URL → 403; anonymous → 302 / 401. UT_011, IT_004–IT_006, ST_002
FR-006	The system shall let a Student see and attempt only the examinations of their own Courses	Security (access control)	High	SO-1	Direct URL to another Course's exam → 403. UT_021, IT_009–IT_011, ST_004
FR-017	The system shall derive remaining time from the server clock only	Security (integrity)	High	SO-2	A changed client clock changes nothing. UT_042, ST_008
FR-025	The system shall withhold results until Faculty publishes	Security (confidentiality)	High	SO-1	Results URL → 403 before publication. UT_059,

Req ID	Requirement (shall...)	Type	Priority	Objective	Acceptance criteria / Test case ref
	them				IT_032, IT_033, ST_012
FR-029	The system shall send no correct-answer data in any response to a Student	Security (confidentiality)	High	SO-1	No `is_correct` in any page or JSON reply. IT_035, ST_013
FR-030	The system shall keep an append-only audit trail of every change	Security (accountability)	Medium	SO-2	One entry per change; none editable. UT_068, IT_036, IT_037, ST_013
NFR-011	Passwords shall be stored only as salted adaptive hashes	Security (credentials)	High	SO-3	Only `pbkdf2_sha256$…` stored. IT_001
NFR-012	The system shall lock an account for 15 minutes after 5 failed sign-ins	Security (credentials)	Medium	SO-3	6th attempt refused. IT_038
NFR-013	Every state-changing request shall carry a CSRF token; all queries go through the ORM; error pages show no internals	Security (web)	High	SO-2, SO-3	POST without token → 403; no raw SQL; generic error page. `portal/test_security.py`
NFR-014	Secrets shall be read from environment variables, never committed	Security (secrets)	High	SO-3	Secret scan clean; app refuses to start without the key. System Validation Report

6. Quality attributes & acceptance tests

Quality attribute	Requirements	How it is accepted	Result (build 2d74453 + Defect 1 fix)
Correctness	FR-001 … FR-030	Every FR passes all its Test Plan cases	119 of 119 cases Pass
Performance	NFR-001, NFR-002, NFR-003	Timed evaluation test; 200-user Locust run	Pass. NFR-002: 100 questions in ≤ 5 s. NFR-001 and NFR-003: 200-user runs with 50,000 questions, p95 1.5–1.9 s in 3 of 4; one run at 3.6 s (System Validation Report §4.1)
Reliability	NFR-004, FR-019, FR-021	Restart test; autosave and resume cases	Pass
Security	NFR-011 … NFR-014 and the FRs of §5.1.2	Test Plan §5.1: automated tests plus manual checks	Pass
Usability & accessibility	NFR-009, FR-018	axe audit, keyboard-only attempt	Pass — 0 AA violations
Portability	NFR-005	Chrome, Edge, Firefox; 360 px	Pass
Maintainability	NFR-010	Policy test and code review	Pass
Compliance	NFR-008	Consent and erasure checks	Pass
Process	NFR-006, NFR-007	GitHub history; traceability gate	NFR-007 Pass; NFR-006 Pass with exceptions recorded in `docs/Deviations.pdf`

Exit criteria for acceptance (`docs/Test_Plan.pdf` §5):

- every high-priority functional requirement implemented and verified;
- every automated test passes, and no S1 or S2 defect is open;
- statement coverage at least 80% and branch coverage at least 70% (measured 88.8% and 72.1%);

- the RTM (§8) shows every test case passed.

Acceptance test suites: the 13 system test cases ST_001 … ST_013, one per use case, plus the Security (Test Plan §5.1), Performance, Accessibility and Portability checks of the System Validation Report.

7. System models and diagrams

7.1 UML use case diagrams

The model has four actors and thirteen use cases. It is drawn as two diagrams, one per side of the examination. `docs/Use_Case_Diagram.pdf` shows all of it at once, and `docs/Use_Case_Flow.pdf` gives each use case's main, alternate and exception flows.

Diagram 1 — Student and Clock. Sitting an examination, from registration to the scorecard. UC-11 and UC-13 have no Student association: they run inside the Student's use cases.

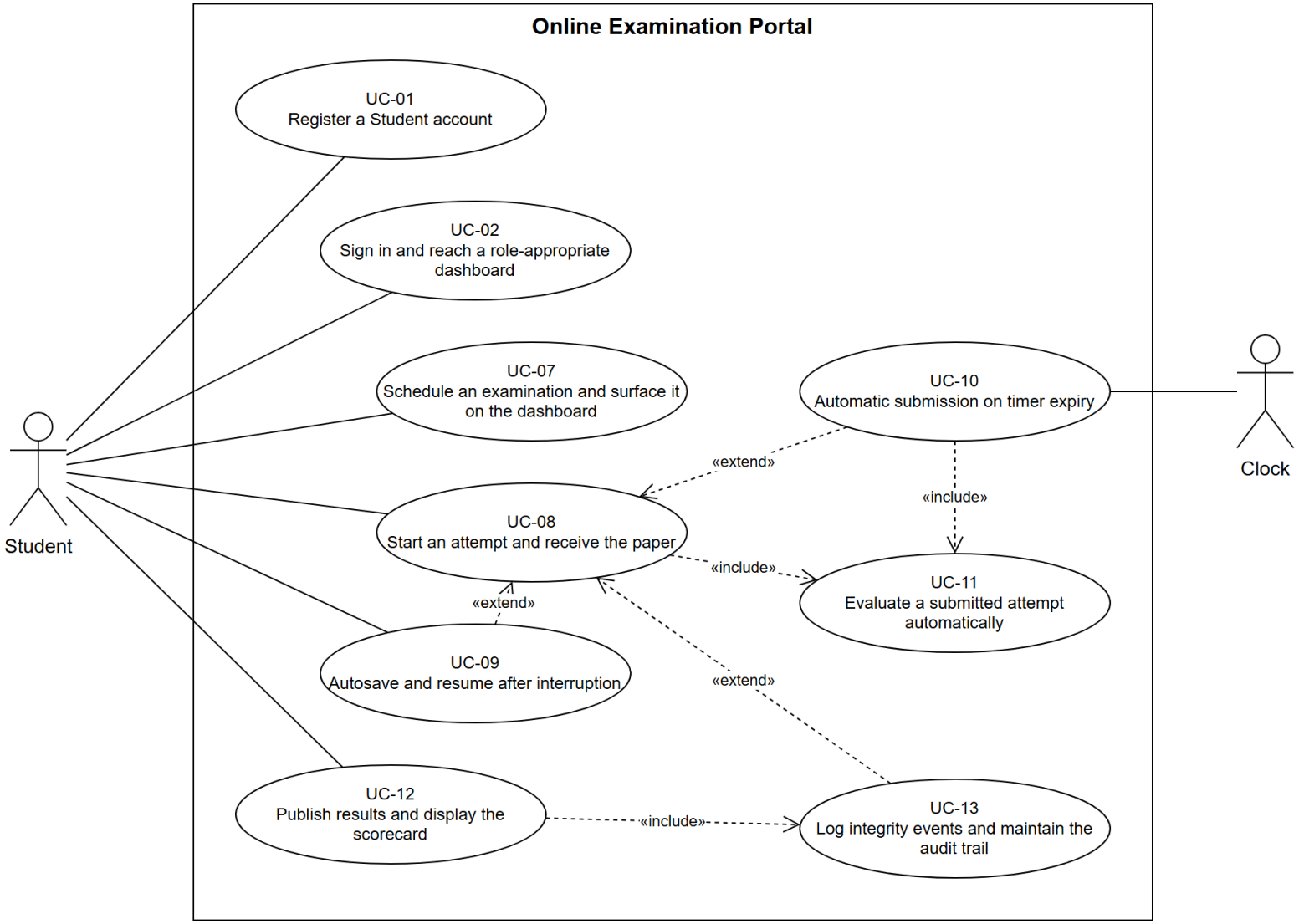
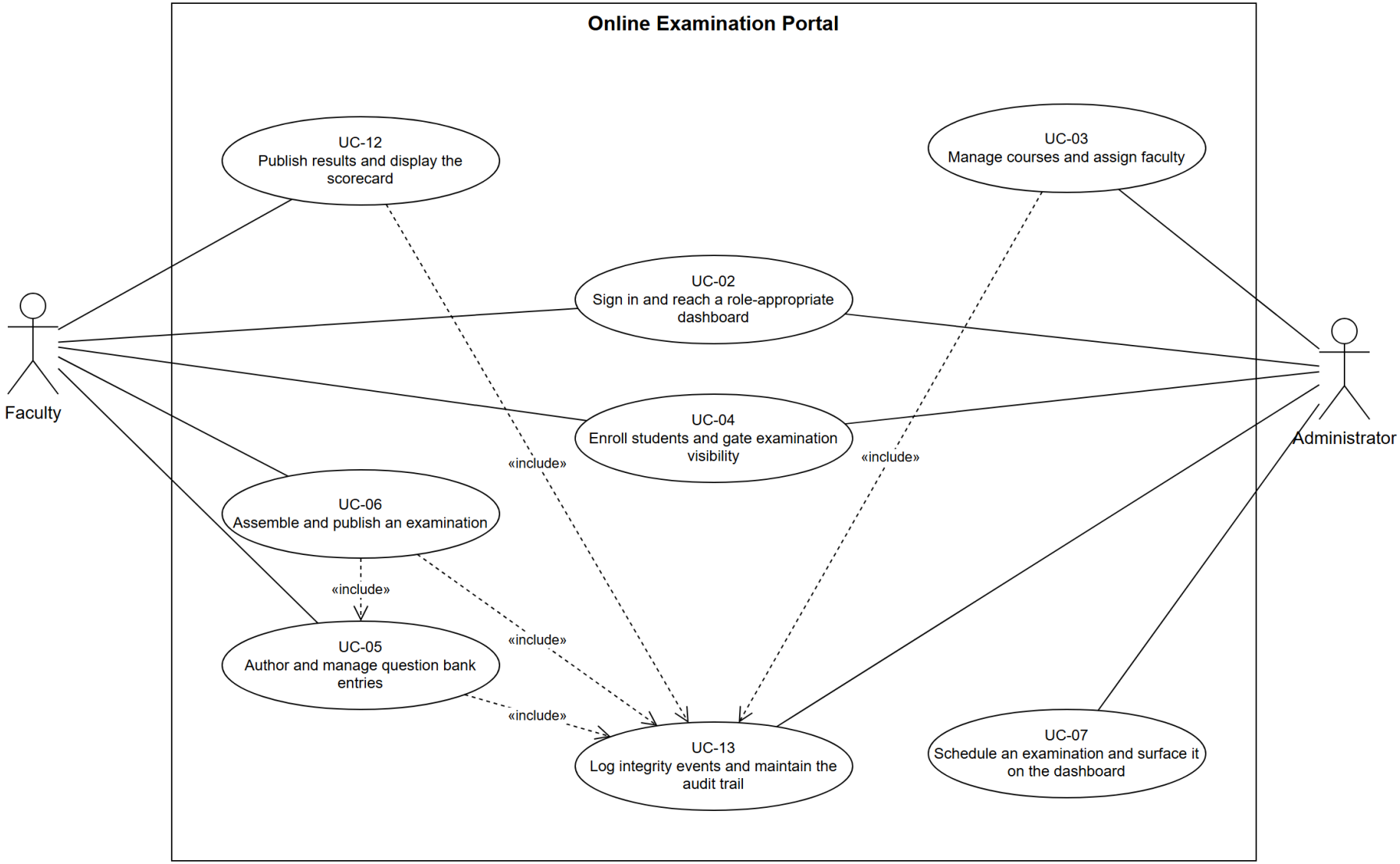


Diagram 2 — Faculty and Administrator. Setting up examinations, releasing results and reviewing the audit trail.


Online Examination Portal
Faculty
Administrator
UC-12
Publish results and display the scorecard
UC-03
Manage courses and assign faculty
UC-02
Sign in and reach a role-appropriate dashboard
UC-04
Enroll students and gate examination visibility
UC-06
Assemble and publish an examination
UC-05
Author and manage question bank entries
UC-13
Log integrity events and maintain the audit trail
UC-07
Schedule an examination and surface it on the dashboard
«include»
«include»
«include»
«include»
«include»

8. Requirements Traceability Matrix (RTM)

Each requirement, where it is specified, the design component and Django app that realise it, the test cases that verify it and its status. The same FR-to-case matrix is held in code (`portal/traceability.py`) and checked by a test, so it cannot silently go stale (NFR-007).

Status: P = passed, all its test cases pass · **A** = accepted with a recorded deviation · **N** = not yet verified. Status is as of build 2d74453 plus the Defect 1 fix (11 Nov) and the load test (12 Nov).

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
FR-001	Register a Student account	4.1 / SAD 3.4 Identity & Enrollment Mgr	`identity`	UT_001–UT_006, IT_001, IT_002, ST_001	P	
FR-002	Sign in	4.1 / SAD 3.4 Identity & Enrollment Mgr	`identity`	UT_007–UT_010, IT_003, ST_002	P	
FR-003	One role, enforced server-side	4.1 / SAD 3.9 RoleMiddleware	`identity`	UT_011, IT_004–IT_006, ST_002	P	Security requirement
FR-004	Manage courses, assign Faculty	4.2 / SAD 3.4 Identity & Enrollment Mgr	`identity`	UT_012–UT_016, IT_007, IT_008, ST_003	P	
FR-005	Enroll Students	4.2 / SAD 3.4 Identity & Enrollment Mgr	`identity`	UT_017–UT_020, ST_004	P	
FR-006	Own Courses' exams only	4.2 / SAD 3.4 IEligibility	`identity`	UT_021, IT_009–IT_011, ST_004	P	Security requirement
FR-007	Author a single-correct MCQ	4.3 / SAD 3.4 Question Bank Mgr	`questionbank`	UT_022–UT_026, ST_005	P	
FR-008	Mandatory question tags and marks	4.3 / SAD 3.4 Question Bank Mgr	`questionbank`	UT_027, IT_012, ST_005	P	
FR-009	Version questions in use	4.3 / SAD 3.4 Question Bank Mgr	`questionbank`	IT_013, IT_014, ST_005	P	
FR-010	Create an Examination with its policy	4.4 / SAD 3.4 Exam Setup Mgr	`examsetup`	UT_028–UT_031, ST_006	P	
FR-011	Add questions from the bank	4.4 / SAD 3.4 IQuestionSource	`examsetup`	IT_015, ST_006	P	
FR-012	Draft, sent to Admin, published (CR-02)	4.4 / SAD 3.4 Exam Setup Mgr	`examsetup`	UT_032, UT_033, IT_016, ST_006	P	
FR-013	Administrator schedules the window (CR-02)	4.5 / SAD 3.4 Exam Setup Mgr	`examsetup`	UT_034, UT_035, IT_017, ST_007	P	
FR-014	Student dashboard	4.5 / SAD 3.4 Attempt Session Mgr	`attempts`	UT_036–UT_038, IT_018, ST_007	P	
FR-015	Start only in the window, once	4.5 / SAD 4.2 Sequence 1	`attempts`	UT_039–UT_041, ST_008	P	
FR-016	Attempt record before the paper	4.6 / SAD 4.2 Sequence 1	`attempts`	IT_019, IT_020, ST_008	P	
FR-017	Server-clock timer	4.6 / SAD 3.9 controls	`attempts`	UT_042, ST_008	P	Security requirement
FR-018	One question at a time, palette	4.6 / SAD 4.5 UX design	`attempts`	UT_043, UT_044, IT_021, ST_008	P	
FR-019	Autosave within 5 s	4.6 / SAD 4.2 Sequence 2	`attempts`	UT_045–UT_047, IT_022, IT_023, ST_009	P	IT_022 run by hand in a browser
FR-020	Auto-submit at zero	4.6 / SAD 4.2 Sequence 2	`attempts`	UT_049–UT_052, IT_026–IT_028, ST_010	P	
FR-021	Resume after interruption	4.6 / SAD 4.2 Sequence 2	`attempts`	UT_048, IT_024, IT_025, ST_009	P	

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
FR-022	Automatic evaluation	4.7 / SAD 3.4 Evaluation Engine	`evaluation`	IT_028–IT_030, ST_011	P	
FR-023	Positive and negative marking	4.7 / SAD 3.4 Evaluation Engine	`evaluation`	UT_053–UT_056, IT_031, ST_011	P	Assumes flat deduction (§2.6)
FR-024	Total, percentage, Pass/Fail	4.7 / SAD 3.4 Evaluation Engine	`evaluation`	UT_057, UT_058, IT_031, ST_011	P	Assumes inclusive pass mark (§2.6)
FR-025	Withhold until published	4.8 / SAD 4.2 Sequence 3	`results`	UT_059, IT_032, IT_033, ST_012	P	Security requirement
FR-026	Publish and unpublish results	4.8 / SAD 4.2 Sequence 3	`results`	UT_060, UT_061, IT_034, ST_012	P	
FR-027	Scorecard	4.8 / SAD 3.4 Results Mgr	`results`	UT_062, UT_063, IT_034, ST_012	P	
FR-028	Record focus loss	4.9 / SAD 3.4 Integrity & Audit Monitor	`integrity`	UT_064–UT_067, ST_013	P	Defect 1 fixed 11 Nov (Deviations D-15)
FR-029	No answer key to the client	4.9 / SAD 3.9 controls	`attempts`	IT_035, ST_013	P	Security requirement
FR-030	Append-only audit trail	4.9 / SAD 3.4 Exam Repository	`repository`, `integrity`	UT_068, IT_036, IT_037, ST_013	P	Security requirement
NFR-001	95% of requests ≤ 2 s at 200 users	5 / SAD 3.6, 3.7	all	`loadtest/locustfile.py`	P	p95 ≤ 2 s in 3 of 4 runs at 200 users; one run 3.6 s with 2 lock timeouts, a known SQLite risk (D-18)
NFR-002	Evaluate 100 questions ≤ 5 s	5 / SAD 3.5	`evaluation`	IT_030	P	
NFR-003	200 attempts, 50,000 questions	5 / SAD 3.7	all	`loadtest/locustfile.py`	P	Every NFR-001 run had the 50,000-question bank seeded
NFR-004	Recover within 60 s after restart	5 / SAD 3.6	`repository`, `attempts`	`portal/test_restart.py`	P	
NFR-005	Browsers and 360 px	5 / SAD 4.5	templates	`portal/test_responsive.py` ; validation	P	
NFR-006	One repo, reviewed PRs	5 / —	process	GitHub history	A	Admin merges recorded in `docs/Deviations.pdf`
NFR-007	Every FR has a test case	5 / —	`portal`	`portal/test_traceability.py`	P	
NFR-008	DPDP consent and erasure	5 / SAD 4.3	`identity`	`identity/test_dpdp.py`; validation	P	
NFR-009	WCAG 2.1 AA	5 / SAD 4.5	templates	`portal/test_accessibility.py`; axe audit	P	
NFR-010	Configurable exam policy	5 / SAD 3.4 Exam Setup Mgr	`examsetup`, `evaluation`	`evaluation/test_policy.py`	P	
NFR-011	Salted adaptive password hashes	5.1.2 / SAD 3.9	`identity`	IT_001	P	Security requirement
NFR-012	Lockout after 5 failures	5.1.2 / SAD 3.9	`identity`	IT_038	P	Security requirement
NFR-013	CSRF, ORM-only, generic errors	5.1.2 / SAD 3.9, 4.4	`portal`, all	`portal/test_security.py`	P	Security requirement
NFR-014	Secrets from the environment	5.1.2 / SAD 3.9	`portal`	Secret scan; start-up check	P	Security requirement

Totals: 43 P · 1 A · 0 N, out of 44 requirements.